# Linear Programming

Graded Assignments 1

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## 1 Modelling

The decision variables for this problem are the amount of red, green and blue sand (in kg) per bucket:

| | $b_1$ | $b_2$ | $b_3$ | $b_4$ | $b_5$ |
|---|---|---|---|---|---|
| kg of red sand | $x_{1,r}$ | $x_{2,r}$ | $x_{3,r}$ | $x_{4,r}$ | $x_{5,r}$ |
| kg of blue sand | $x_{1,b}$ | $x_{2,b}$ | $x_{3,b}$ | $x_{4,b}$ | $x_{5,b}$ |
| kg of green sand | $x_{1,g}$ | $x_{2,g}$ | $x_{3,g}$ | $x_{4,g}$ | $x_{5,g}$ |

For example, $x_{2,b}$ represents the amount of blue sand in bucket 2 (in kg), etc.

### Objective Function

The goal of the LP is to **minimize** the following function:

$$Z = (x_{1,r}+x_{2,r}+x_{3,r}+x_{4,r}+x_{5,r})\cdot 2.5+(x_{1,b}+x_{2,b}+x_{3,b}+x_{4,b}+x_{5,b})\cdot 2+(x_{1,g}+x_{2,g}+x_{3,g}+x_{4,g}+x_{5,g})$$

### Constraints

Our supplier only has a limited amount of sand available:

$$x_{1,r} + x_{2,r} + x_{3,r} + x_{4,r} + x_{5,r} \leq 30$$

$$x_{1,b} + x_{2,b} + x_{3,b} + x_{4,b} + x_{5,b} \leq 20$$

$$x_{1,g} + x_{2,g} + x_{3,g} + x_{4,g} + x_{5,g} \leq 25$$

No bucket should be overloaded beyond its maximum capacity:

$$x_{1,r} + x_{1,b} + x_{1,g} \leq 7$$

$$x_{2,r} + x_{2,b} + x_{2,g} \leq 8$$

$$x_{3,r} + x_{3,b} + x_{3,g} \leq 5$$

$$x_{4,r} + x_{4,b} + x_{4,g} \leq 20$$

$$x_{5,r} + x_{5,b} + x_{5,g} \leq 15$$

Each bucket should be at least half-filled:

$$x_{1,r} + x_{1,b} + x_{1,g} \geq 3.5$$

$$x_{2,r} + x_{2,b} + x_{2,g} \geq 4$$

$$x_{3,r} + x_{3,b} + x_{3,g} \geq 2.5$$

$$x_{4,r} + x_{4,b} + x_{4,g} \geq 10$$

$$x_{5,r} + x_{5,b} + x_{5,g} \geq 7.5$$

In each bucket, the amount of red sand used should be at least half of the total amount of sand in the bucket:

$$x_{1,r} \geq x_{1,b} + x_{1,g}$$

$$x_{2,r} \geq x_{2,b} + x_{2,g}$$

$$x_{3,r} \geq x_{3,b} + x_{3,g}$$

$$x_{4,r} \geq x_{4,b} + x_{4,g}$$

$$x_{5,r} \geq x_{5,b} + x_{5,g}$$

For each two adjacent buckets, the total amount of blue sand in the two buckets should be at least 5kg:

$$x_{1,b} + x_{2,b} \geq 5$$

$$x_{2,b} + x_{3,b} \geq 5$$

$$x_{3,b} + x_{4,b} \geq 5$$

$$x_{4,b} + x_{5,b} \geq 5$$

Lastly, a bucket cannot have a negative amount of sand in it (nonnegativity constraints):

$$x_{i,j} \geq 0, \text{ for all } i = \{1, 2, 3, 4, 5\}, j = \{\mathrm{r}, \mathrm{b}, \mathrm{g}\}$$

# 2 Simplex Method

Consider the following linear program:

$$
\begin{array}{lrcrcrcrcrcrcl}
\max & Z = 3x_1 + x_2 + 2x_3 & & & & & & & & & & & & \\
\text{s.t.} & -x_1 & + & 2x_2 & - & x_3 & + & x_4 & & & & & = & 20 \\
 & 2x_1 & - & 4x_2 & + & 2x_3 & & & + & x_5 & & & = & 30 \\
 & 3x_1 & + & 3x_2 & + & x_3 & & & & & + & x_6 & = & 25 \\
\text{and} & x_j \geq 0 & & & & & & & & & & & &
\end{array}
$$

Where the red-colored variables $x_4, x_5, x_6$ are slack variables.

## 2.1 Solve this LP using the Simplex method

We can transform the above LP into the following tableau:

| | | $Z$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | RHS |
|---|---|---|---|---|---|---|---|---|---|
| $z$ | (0) | 1 | $-3$ | $-1$ | $-2$ | 0 | 0 | 0 | 0 |
| $x_4$ | (1) | 0 | $-1$ | 2 | $-1$ | 1 | 0 | 0 | 20 |
| $x_5$ | (2) | 0 | 2 | $-4$ | 2 | 0 | 1 | 0 | 30 |
| $x_6$ | (3) | 0 | 3 | 3 | 1 | 0 | 0 | 1 | 25 |

Here $x_1$ is the entering variable, as -3 is the most negative value in the top row. We can find the leaving variable by applying the minimum ratio test:

- row (2): $\frac{30}{2} = 15$
- row (3): $\frac{25}{3} = 8\frac{1}{3}$

$8\frac{1}{3} < 15$, thus $x_6$ (row (3)) is the leaving variable.

By applying Gaussian elimination, we reach the following tableau:

| | | $Z$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | RHS |
|---|---|---|---|---|---|---|---|---|---|
| $z$ | (0) | 1 | 0 | 2 | $-1$ | 0 | 0 | 1 | 25 |
| $x_4$ | (1) | 0 | 0 | 3 | $-\frac{2}{3}$ | 1 | 0 | $\frac{1}{3}$ | $28\frac{1}{3}$ |
| $x_5$ | (2) | 0 | 0 | $-6$ | $1\frac{1}{3}$ | 0 | 1 | $-\frac{2}{3}$ | $13\frac{1}{3}$ |
| $x_1$ | (3) | 0 | 1 | 1 | $\frac{1}{3}$ | 0 | 0 | $\frac{1}{3}$ | $8\frac{1}{3}$ |

Here $x_3$ is the entering variable, as -1 is the most negative value in the top row. We can find the leaving variable by applying the minimum ratio test:

- row (2): $\frac{13\frac{1}{3}}{1\frac{1}{3}} = 10$
- row (3): $\frac{8\frac{1}{3}}{\frac{1}{3}} = 25$

$10 < 25$, thus $x_5$ (row (2)) is the leaving variable.

By applying Gaussian elimination, we reach the following tableau:

| | | $Z$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | RHS |
|---|---|---|---|---|---|---|---|---|---|
| $z$ | (0) | 1 | 0 | $-2\frac{1}{2}$ | 0 | 0 | $\frac{3}{4}$ | $\frac{1}{2}$ | 35 |
| $x_4$ | (1) | 0 | 0 | 0 | 0 | 1 | $\frac{1}{2}$ | 0 | 35 |
| $x_3$ | (2) | 0 | 0 | $-4\frac{1}{2}$ | 1 | 0 | $\frac{3}{4}$ | $-\frac{1}{2}$ | 10 |
| $x_1$ | (3) | 0 | 1 | $2\frac{1}{2}$ | 0 | 0 | $-\frac{1}{4}$ | $\frac{1}{2}$ | 5 |

Here $x_2$ is the entering variable, as $-2\frac{1}{2}$ is the most negative value in the top row. $x_1$ is the leaving variable, as $2\frac{1}{2}$ is the only postive non-zero element in the column of $x_2$.

By applying Gaussian elimination, we reach the following tableau:

| | | $Z$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | RHS |
|---|---|---|---|---|---|---|---|---|---|
| $z$ | (0) | 1 | $1\frac{2}{3}$ | 0 | 0 | 0 | $\frac{1}{3}$ | $1\frac{1}{3}$ | 40 |
| $x_4$ | (1) | 0 | 0 | 0 | 0 | 1 | $\frac{1}{2}$ | 0 | 35 |
| $x_3$ | (2) | 0 | 3 | 0 | 1 | 0 | 0 | 1 | 19 |
| $x_2$ | (3) | 0 | $\frac{2}{3}$ | 1 | 0 | 0 | $-\frac{1}{6}$ | $\frac{1}{3}$ | 2 |

Now we have reached optimality, as there are no negative values in the top row. The value of the objective function is:

$$Z = 40$$

And the variable assignment is:

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 2, 19, 35, 0, 0)$$

## 2.2 Which variables are (non)basic?

$\{x_2, x_3, x_4\}$ is the basis, so $x_2$, $x_3$ and $x_4$ are basic variables. The other variables, $\{x_1, x_5, x_6\}$, are nonbasic variables.

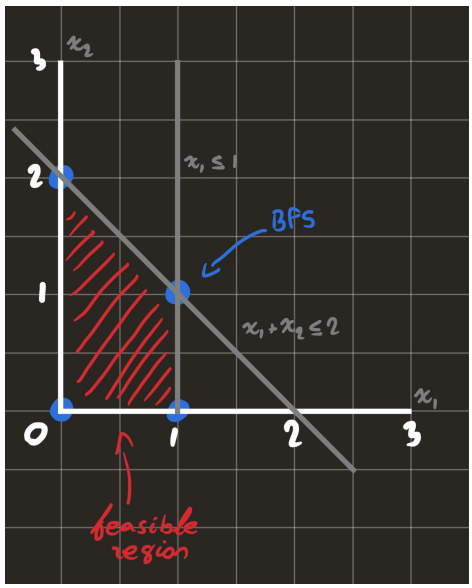
# 3 Degeneracy

We are given a linear program with two non-negative decision variables, $x_1$ and $x_2$, and two functional constraints:

$$x_1 \leq 1$$
$$x_1 + x_2 \leq 2$$

## 3.1 List all the basic feasible solutions of this program

We can visualize this program as follows:




The basic feasible solutions are represented by the (blue-colored) corner points of the feasible region. If we rewrite the constraints to include slack variables:

$$x_1 + x_3 = 1$$
$$x_1 + x_2 + x_4 = 2$$

We can read off all the basic feasible solutions, with their respective basis' and tight constraints as follows:

- $(x_1, x_2, x_3, x_4) = (0, 0, 1, 2)$, with basis $\{x_3, x_4\}$. The tight constraints here are the non-negativity constraints of $x_1$ and $x_2$.
- $(x_1, x_2, x_3, x_4) = (0, 2, 1, 0)$, with basis $\{x_2, x_3\}$. The tight constraints here are the non-negativity constraint of $x_1$ and $x_1 + x_2 \leq 2$.
- $(x_1, x_2, x_3, x_4) = (1, 0, 0, 1)$, with basis $\{x_1, x_4\}$. The tight constraints here are the non-negativity constraint of $x_2$ and $x_1 \leq 1$.
- $(x_1, x_2, x_3, x_4) = (1, 1, 0, 0)$, with basis $\{x_1, x_2\}$. The tight constraints here are $x_1 \leq 1$ and $x_1 + x_2 \leq 2$.

## 3.2 How many basic solutions does this program have?

The basic solutions of this program are represented by *all* corner points of the visualization above. This includes the basic feasible solutions shown above, as well as the corner point $(x_1, x_2, x_3, x_4) = (2, 0, -1, 0)$. This makes for a total of 5 basic solutions.

## 3.3 Add a third constraint such that the resulting linear program has at least one degenerate BFS

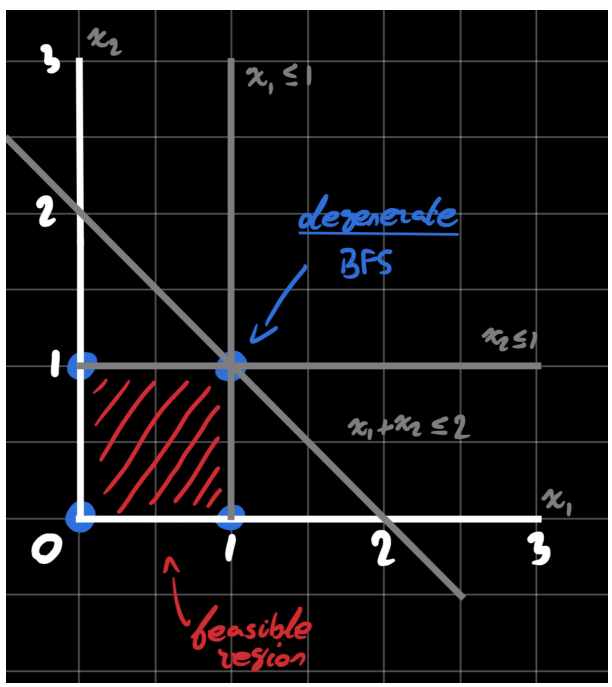
One way to do this is to introduce a new constraint boundary that intercepts the other constraint boundaries at the same BFS. This BFS will then become degenerate. The constraint we will introduce is:

$$x_2 \leq 1$$

Which can also be written as:

$$x_2 + x_5 = 1$$

We can visualize this program as follows:




The BFS at the intersection of the constraint boundaries represents the solution $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 0, 0, 0)$. This BFS is degenerate, because it has a basic variable that is equal to 0.